HEALTH-A-THON 2026 · CANCER TRACK · PATIENT / CAREGIVER

# EKYATRA Saathi

Readiness-driven visit planning that cuts wasted trips for families travelling for cancer care.

[Team name] · Use case: Patient Follow-up & Continuity of Care

THE PROBLEM

# Cancer care means a long journey, and the journey itself is ruinous

**60%**

of Tata Memorial Hospital's ~45,000 new patients a year come from outside Maharashtra **(1)**

**75%+**

of gastric and pancreatic cancer patients studied at Tata Memorial Centre travelled over 500 km **(2)**

**83%**

faced catastrophic **non-medical** costs. Mean spend: ₹88,433 on travel, lodging and food **(2)**

> Travel (43%) and accommodation (33%) make up most of that cost. Among patients travelling over 500 km, 53% borrowed or sold assets to cope **(2)**.

(1) C.S. Pramesh, Director, TMH, in Think Global Health, Oct 2022 · (2) IIPS and Tata Memorial Centre, Journal of Cancer Policy, 2025

WHY IT PERSISTS

# Nobody counts the family's trips

Every department optimises its own queue. The caregiver holds the whole journey together alone, often in a language the hospital does not speak.

> **90%** of colorectal cancer patients across five tertiary hospitals in four states faced catastrophic spending. **9.4%** had an unplanned treatment interruption **(3)**.

| | |
|---|---|
| **Pathology lab** | Report turnaround |
| **Radiology** | Scan slots |
| **OPD** | Consult queue |
| **The family** | Trips, days, wages: unmeasured |

(3) CROCODILE study group, The Lancet Regional Health: Southeast Asia, 2022 · Families camping on footpaths outside Tata Memorial: PARI, Mar 2020

THE SOLUTION

# One rule: never travel for a step that is not ready

01

### Read status

Appointment and report status from ABDM records and hospital feeds. Never report content.

02

### Check what each visit needs

A dependency graph blocks any visit whose prerequisite is not ready. Uncertain? No date is promised.

03

### Tell the caregiver

WhatsApp or voice call in their language: whether to come, what to bring, where to go first.

04

### Notice the silence

When a family goes quiet after a visit, a hospital navigator or support NGO is alerted.

**What is different:** it optimises the family's trips and days away from home, not any department's throughput.

ONE FAMILY'S JOURNEY

# Same care, fewer trips

**Today**
4 trips

- **Trip 1** — Consult and biopsy
- **Trip 2: wasted** — Report not ready
- **Trip 3** — Scan
- **Trip 4** — Review consult

**With Saathi**
2 trips

- **Trip 1** — Consult, biopsy and bloods in one visit
- **At home** — Voice message in Bengali: "Report is ready. Come Thursday."
- **Trip 2** — Scan and review consult on the same trip, documents ready

Illustrative journey for the prototype demo, run on synthetic patient data.

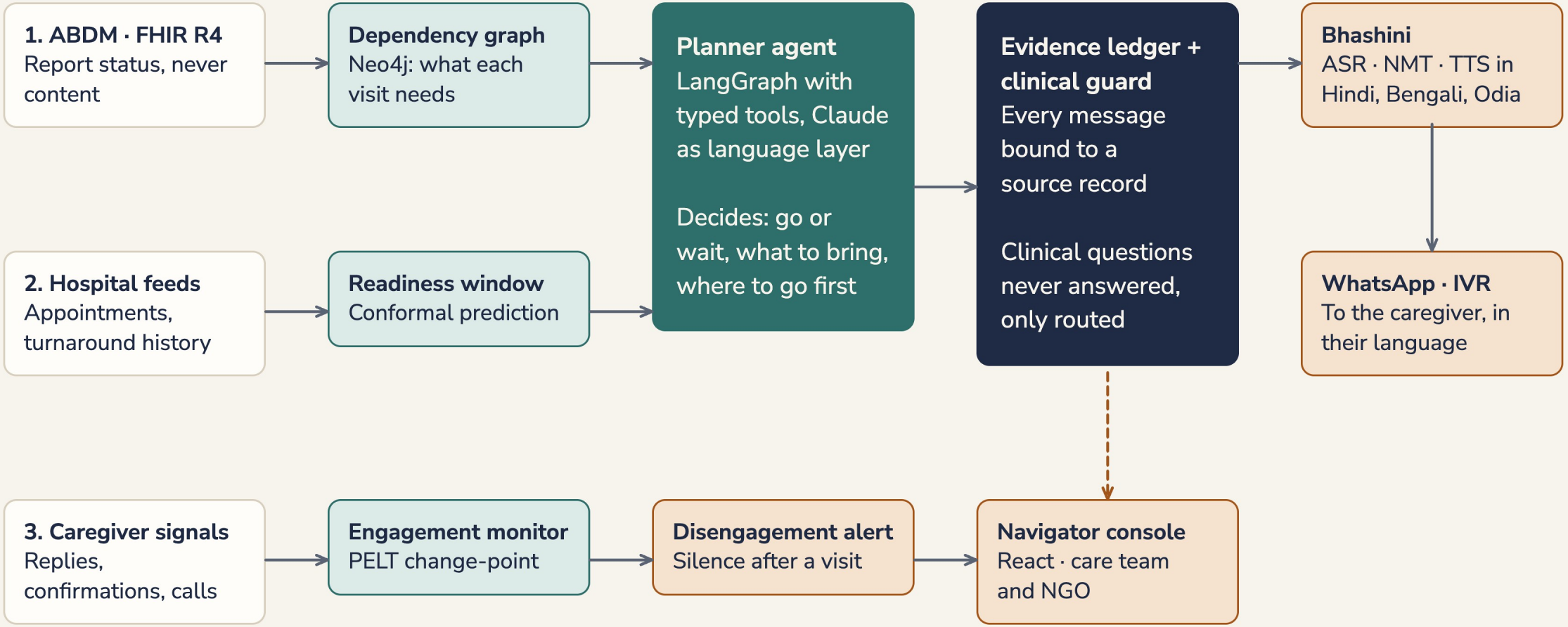
HOW WE BUILD IT

# Three layers, every message traceable

| | |
|---|---|
| **Reach the family** | WhatsApp Business API · Exotel IVR voice calls · Bhashini ASR, translation and TTS for Hindi, Bengali and Odia |
| **Plan the visit** | LangGraph agent with typed tools · Neo4j dependency graph · conformal prediction for readiness windows · PELT change-point for disengagement alerts · Claude as language layer, bound to an evidence ledger |
| **Read status** | ABDM and FHIR R4 APIs · hospital appointment feeds · FastAPI and PostgreSQL · React console for navigators |

TECHNICAL ARCHITECTURE

# Status in, a safe message out



SCOPE AND IMPACT

# Logistics only, measured in the family's terms

## It never

- Diagnoses or recommends treatment
- Reads or interprets report content
- Scores clinical risk
- Answers a clinical question: it routes it to the care team

## We measure

- Wasted trips avoided
- Days the family spends in the city
- Non-medical out-of-pocket cost avoided (₹)
- Dropout after the first visit

Prototype: synthetic patients with planted ground truth · Pilot: live ABDM data and one hospital's status feed